Sudden oak death

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1 Overview

Sudden oak death, caused by the water mould *Phytophthora ramorum*, has killed millions of tanoaks and oaks in coastal California and Oregon since the mid-1990s. Its spores are splashed and blown from infected foliage, mostly a few metres, occasionally kilometres, and only in wet, mild weather. Meentemeyer et al. (2011) modelled its spread across California from 1990 to 2030 on a grid of 250 m cells, each holding up to 100 units of host vegetation, susceptible or infected. Infected units release spores when the weather and the season allow; the spores travel by a mixture of a short-range and a long-range dispersal kernel, and infect susceptible units where they land. They predicted a ten-fold increase in infection between 2010 and 2030, much more in favourable weather, and little benefit from curbing long-distance spread once the disease is established.

This implementation is an adaptation. The source's statewide maps of host vegetation and weekly weather were never published, so it runs on a synthetic landscape with synthetic weather, and reproduces the source's qualitative results rather than its areas.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Statistics
```

2 Dispersal

A spore travels in a random direction a distance drawn from a Cauchy kernel, with scale 20.57 m with probability γ = `0.9947` and 9.5 km otherwise. On cells of 250 m, a spore released at a cell's centre lands in each cell with probability given by `sod_kernel`:

```
short = sod_kernel(20.57; radius=4)
long = sod_kernel(9504; radius=99)
(short_in_own_cell=round(short[5, 5]; digits=3),
 short_within_1km=round(sum(short); digits=3),
 long_within_12km=round(sum(long[(100 - 49):(100 + 49), (100 - 49):(100 + 49)]);
                        digits=2))
```

```
(short_in_own_cell = 0.906, short_within_1km = 0.989, long_within_12km = 0.62)
```

As the source says, nearly all spores stay close: about nine in ten short-range spores land in their own cell (93% in the source). The rare long-range spores spread over tens of kilometres.

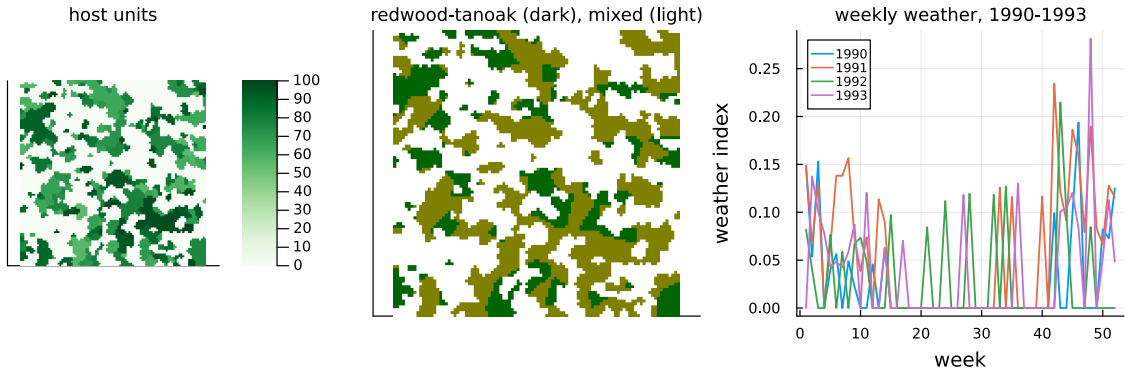
Because each infected unit releases a Poisson number of spores and each lands independently, the spores landing in a cell are Poisson, with a mean that is the sum over all infected cells of their spores times the probability of landing there: a convolution. The model computes it as two kernel fields of the core package, the long-range one by fast Fourier transform, on a torus twice the landscape's size whose padding is masked, so spores leaving the landscape are lost.

3 Landscape and weather

The synthetic landscape is 100×100 cells (25×25 km). Half the cells hold host vegetation, in clusters; a third of the forest is redwood-tanoak, the rest mixed evergreen. Redwood-tanoak forest takes part in transmission from January to September, mixed evergreen from February to September.

```
prob = problem(:sudden_oak_death; seed=1)
n = 100
hosts = prob.u0.S[1:n, 1:n] .+ prob.u0.I[1:n, 1:n]
p1 = heatmap(Float64.(hosts); c=:Greens, aspect_ratio=1, title="host units",
             titlefontsize=10, xticks=false, yticks=false)
p2 = heatmap(Float64.(prob.u0.forest[1:n, 1:n]);
             c=cgrad([:white, :darkgreen, :olive], 3; categorical=true),
aspect_ratio=1,
```

```
            title="redwood-tanoak (dark), mixed (light)", titlefontsize=10,
            xticks=false, yticks=false, colorbar=false)
w = prob.p.weather
p3 = plot(1:52, reshape(w[1:(52 * 4)], 52, 4); lw=1.5, xlabel="week",
          ylabel="weather index", label=permutedims(string.(1990:1993)),
          title="weekly weather, 1990-1993", titlefontsize=10, legendfontsize=7)
plot(p1, p2, p3; layout=(1, 3), size=(950, 300), left_margin=3Plots.mm,
     bottom_margin=4Plots.mm)
```



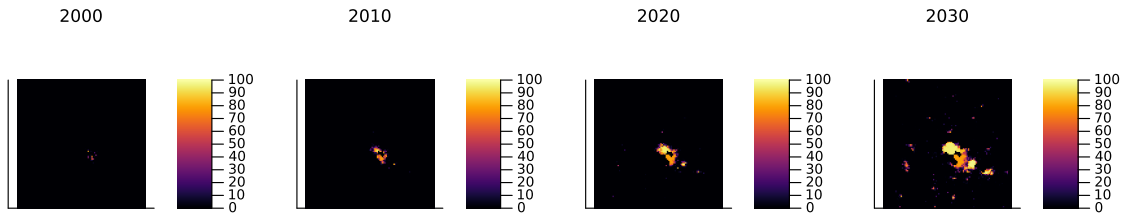
Each week's weather index is the share of its days with rain times the source's suitability of its mean temperature for sporulation; it enters both at the source of the spores and where they land. The source used kriged weather at 250 m for 1990 to 2008. Here a coastal climate is drawn at random: wet, mild winters and springs, dry summers, and wetter and drier years. Its overall wetness is the one choice made to match the source: with it, infection grows about ten-fold from 2010 to 2030.

4 An epidemic, 1990 to 2030

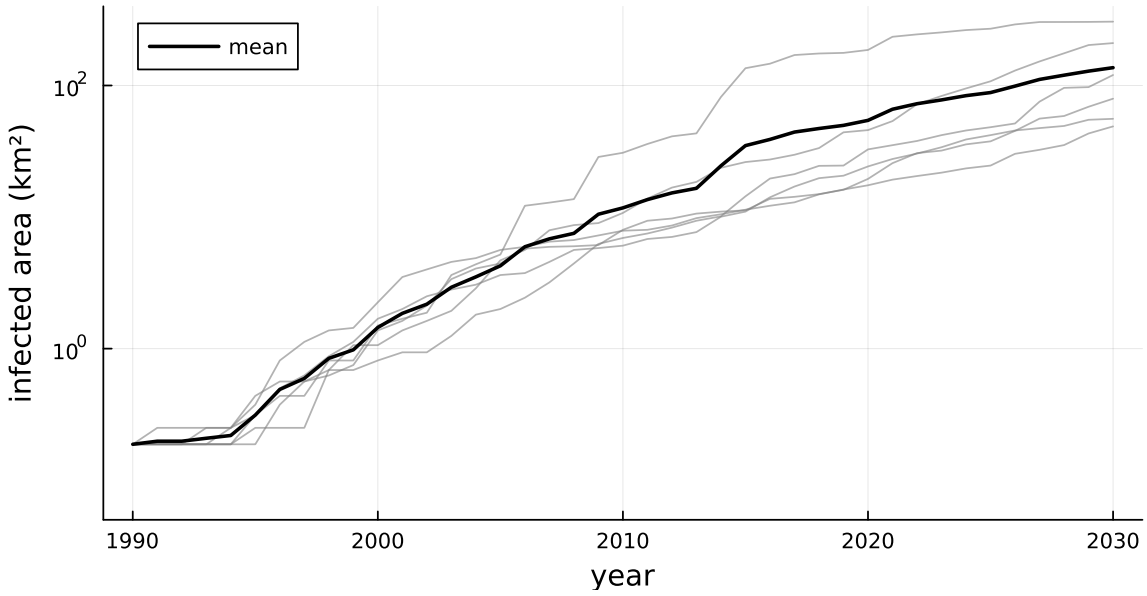
The pathogen arrives at three cells near the centre in 1990, each with one infected unit. Six runs:

```
function epidemic(; snapshots=(), kwargs...)
    integ = init(problem(:sudden_oak_death; kwargs...); save_everystep=false)
    area = Float64[]
    maps = Dict{Int,Matrix{Float64}}()
    for year in 1990:2030
        for _ in 1:52
            step!(integ)
        end
        push!(area, sod_infected_area(integ.u))
        year in snapshots && (maps[year] = Float64.(integ.u.I[1:n, 1:n]))
    end
    return area, maps
end
ensemble(; seeds=1:6, kwargs...) =
```

```
    fetch.([Threads.@spawn first(epidemic(; seed=s, kwargs...)) for s in seeds])
_, maps = epidemic(; seed=1, snapshots=(2000, 2010, 2020, 2030))
plot([heatmap(maps[y]; c=:inferno, clims=(0, 100), aspect_ratio=1, title="$y",
              titlefontsize=10, xticks=false, yticks=false)
      for y in (2000, 2010, 2020, 2030)]...; layout=(1, 4), size=(1000, 230))
```




```
random = ensemble()
A = reduce(hcat, random)
p = plot(1990:2030, A; color=:gray, alpha=0.6, label="", xlabel="year",
         ylabel="infected area (km²)", yscale=:log10, ylims=(0.05, 400))
plot!(p, 1990:2030, mean(A; dims=2); color=:black, lw=2, label="mean", size=(650,
340))
```




```
(area_2010=round(mean(A[21, :]); digits=1), area_2030=round(mean(A[41, :]);
digits=1),
 increase=round(mean(A[41, :]) / mean(A[21, :]); digits=1))
```

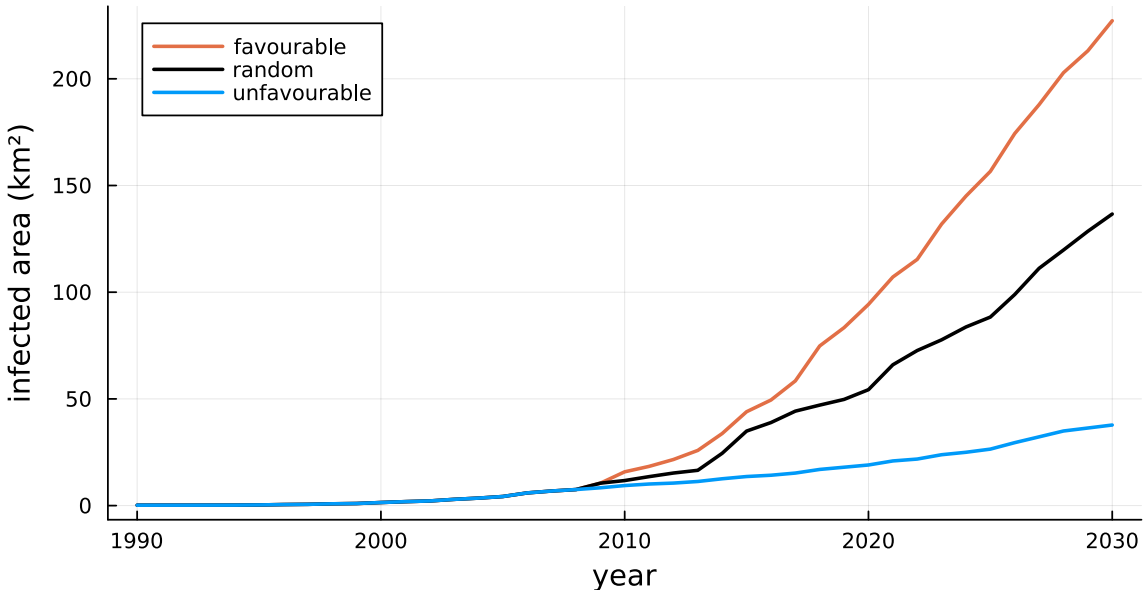
```
(area_2010 = 11.7, area_2030 = 136.6, increase = 11.6)
```

Infected units multiply locally, cell by cell, while the rare long-range spores start new foci kilometres away, which then grow in turn. The infected area grows about ten-fold from 2010 to 2030, as in the source, though here over a landscape of 300 km² of forest rather than California's.

5 Weather scenarios

The source ranks its observed years by their total weather index and draws the years after 2008 at random from the better half (favourable), the worse half (unfavourable), or all of them (random). Six runs each:

```
scenarios = Dict(w => reduce(hcat, ensemble(; weather=w))
                 for w in (:favourable, :random, :unfavourable))
p = plot(; xlabel="year", ylabel="infected area (km²)", size=(650, 340),
legend=:topleft)
for (w, c) in ((:favourable, 2), (:random, :black), (:unfavourable, 1))
    plot!(p, 1990:2030, mean(scenarios[w]; dims=2); lw=2, color=c, label=string(w))
end
p
```




```
(favourable=round(mean(scenarios[:favourable][41, :]); digits=1),
 random=round(mean(scenarios[:random][41, :]); digits=1),
 unfavourable=round(mean(scenarios[:unfavourable][41, :]); digits=1))
```

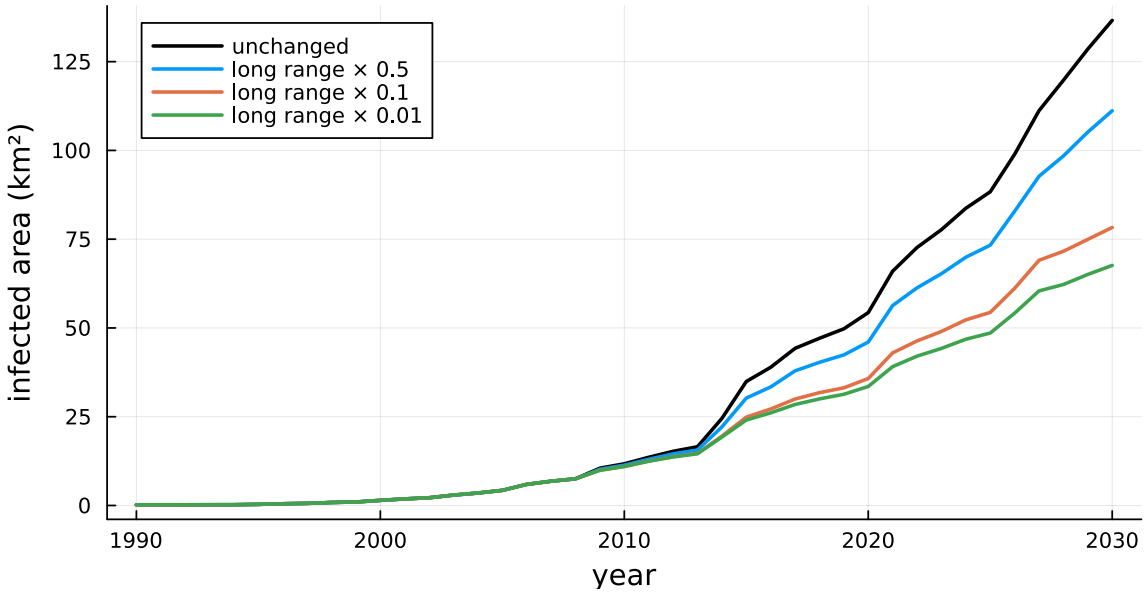
```
(favourable = 227.2, random = 136.6, unfavourable = 37.8)
```

As in the source, the weather after 2008 makes a large difference by 2030: favourable weather gives several times the infection of unfavourable (about five-fold in the source), and random weather a little over half that of favourable (half in the source).

6 Curbing long-distance spread

The source asks what reducing long-distance dispersal from 2009 would achieve, for example by stopping the movement of infected plants. Multiplying the long-range share of spores by 0.5, 0.1 or 0.01:

```
curbs = Dict(f => reduce(hcat, ensemble(; management=(2009, f))) for f in (0.5, 0.1,
0.01))
p = plot(1990:2030, mean(scenarios[:random]; dims=2); lw=2, color=:black,
         label="unchanged", xlabel="year", ylabel="infected area (km²)", size=(650,
340),
         legend=:topleft)
for (k, f) in enumerate((0.5, 0.1, 0.01))
    plot!(p, 1990:2030, mean(curbs[f]; dims=2); lw=2, color=k, label="long range ×
    $f")
end
p
```




```
(unchanged=round(mean(scenarios[:random][41, :]); digits=1),
 curbed=[round(mean(curbs[f][41, :]); digits=1) for f in (0.5, 0.1, 0.01)])
```

```
(unchanged = 136.6, curbed = [111.2, 78.3, 67.6])
```

Curbing long-distance spread slows the epidemic but does not stop it: the foci already established keep growing and merging by short-range spread, and even a hundred-fold cut leaves about half the infected area by 2030. The source finds a smaller benefit, about a third less.

7 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:sudden_oak_death).deviations)
```

```
- The source's statewide host-index, forest-type and weekly weather rasters at 250 m
were never published, so the model runs on a synthetic landscape of 100 x 100 cells
(25 x 25 km): host units from a random cluster landscape on half the cells, a third
of them redwood-tanoak forest and the rest mixed evergreen. Its statewide results
(infected areas in km², invasion probabilities by ecoregion, validation statistics)
cannot be reproduced.
- Weather is synthetic and the same across the landscape: each week's index is the
share of rainy days times the source's temperature suitability, from a seasonal
cycle of rain and temperature with wetter and drier years. The overall wetness (0.5)
was chosen so that infection grows about ten-fold from 2010 to 2030, the source's
headline result; with wetness 1 the landscape is infected by about 2010.
- The source releases spore units as particles. Since emissions are Poisson and each
lands independently, the spore units landing in a cell are Poisson with a mean given
by a convolution of the infected units with the landing probabilities; this is
computed as two kernel fields (short and long range). New infections in a week are
Poisson, capped at the cell's susceptible units at the start of the week; in the
source, each landing spore unit meets the units left by those before it.
- The kernel's mixture weight gamma = 0.9947 is read as the share of spore units
dispersing at short range, each range with a normalised half-Cauchy distance, which
gives the source's 'over 99%' short-range dispersal and about 93% of it within the
source cell (91% here). Long-range spores travelling beyond the landscape's width
are lost, as are those leaving the landscape.
- The source scales spore units so that one spore unit is at most one infection and
retains them by the ratio of actual to maximal infectivity and susceptibility,
described only in words; here an infected unit releases spore units at rate beta w
and a spore unit infects with probability w S / nmax, with w the week's weather
index at both source and destination, as in Eq. 1.
- Weeks are assigned to months by week w lying in month ceil(12 w / 52) for the
seasons of the forest types; the source does not say how.
- The infection starts from one infected unit in each of three host cells near the
landscape's centre; the source's three introduction sites and their initial loads
are not given.
- Management multiplies the long-range share of spore units by a factor from a given
year, removing the rest; the source does not say whether they were moved to short
range instead.
```

Meentemeyer, Ross K., Nik J. Cunniffe, Alex R. Cook, et al. 2011. “Epidemiological Modeling of Invasion in Heterogeneous Landscapes: Spread of Sudden Oak Death in California (1990–2030).” *Ecosphere* 2 (2): art17. https://doi.org/10.1890/ES10-00192.1.